SAMARTH

CUET PG EXAM





Lecture -4 | Computer Science

Digital Fundamental

Topic : Floating Point Representation



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Recap *of previous lecture*

1. Complements



Topics *to be covered*

1. **Floating Point Representation**

Two approaches to represent real nos in computer

① fixed pt

② floating pt



Topic : Floating Point Representation

Fixed pt

Ex 18.25

Decimal point is placed at fixed position

Decimal point can float (move)

Topic : Floating Point Representation

Used to Rep. Very large / very small nos. Ex Age of universe, size of molecule, speed of light, 3×10^8

- The Floating point representation is a way to the encode numbers in a format that can handle very large and very small values. It is based on scientific notation where numbers are represented as a fraction and an exponent. In computing, this representation allows for trade-off between range and precision.

Scientific Notation

$$\pm M \times B^{E}$$

M – Mantissa
B – Base
E – Exponent

		Mantissa	Base	Exponent	
①	3×10^{8}	3	10	8	
②	6183.986	6183986	10	-3	6183986×10^{-3}
③	72.25	7225	10	-2	7225×10^{-2}

Topic : Floating Point Representation

Format: A floating point number is typically represented as:

Value=Sign × Significand × BaseExponent

- Sign: Indicates whether the number is positive or negative.
- Significand (Mantissa): Represents the precision bits of the number.
- Base: Usually 2 in binary systems.
- Exponent: Determines the scale of the number.

Topic : Floating Point Representation

IEEE 754 Floating pt Rep Precision format.

- In computer systems the floating-point numbers serve a very important purpose especially in scientific computing, computer graphics, and machine learning as it is used to represent real numbers. These floating-point numbers can be defined with different precision formats which are main single precision and double precision.

Q Direct Expected

Precision	Base	Sign	Exponent	Significand
① Single precision	2	1	8	23+1
② Double precision	2	1	11	52+1

Format & no of bits

Topic : Floating Point Representation

- The two forms of Single and Double Precision really and truly differs by the number of bits that are used to represent these numbers and so varies the range and also the accuracy of the amount of numbers that can be found. Single precision requires 32 bits on a binary format while the double precision requires 64 bits. That is why it is critical to understand the difference in cases when the choice has to be made in favor of either speed or accuracy when it comes to computations.

Topic : Floating Point Representation

- Single Precision : a format proposed by IEEE for the representation of floating-point numbers. It occupies 32 bits in computer memory.

LEARN.



Sign bit

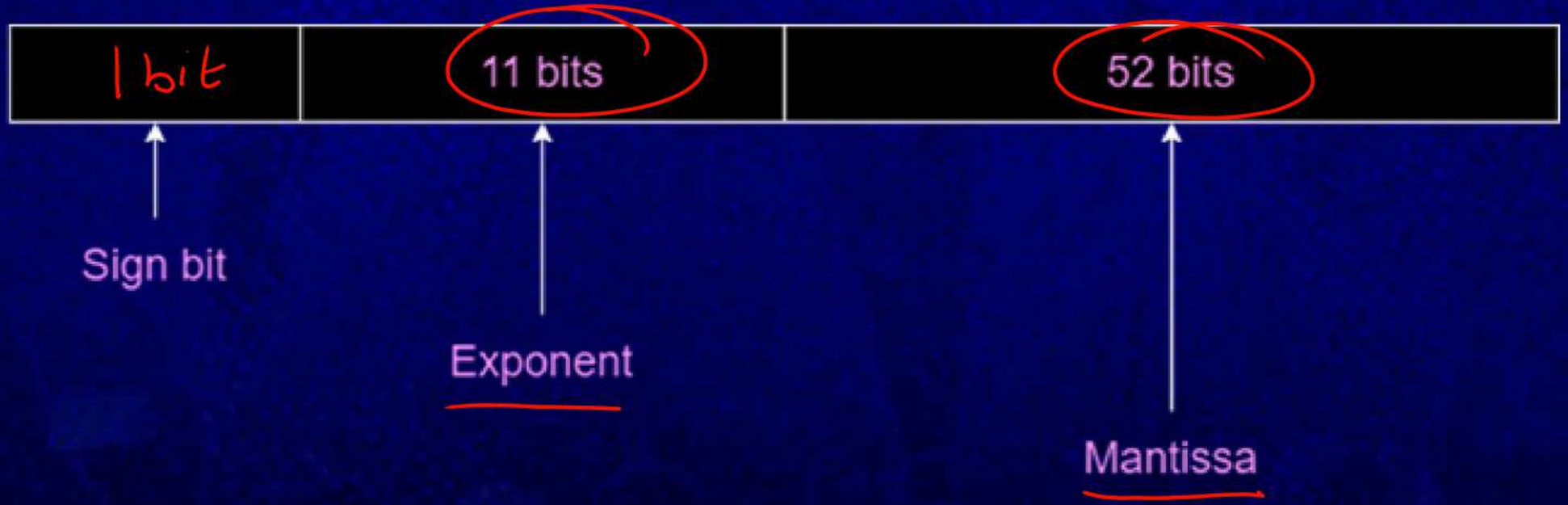
Exponent

Mantissa

0 - Positive
1 - Negative

Topic : Floating Point Representation

Double Precision: It occupies 64 bits in computer memory.



Ex Represent $(1259.125)_{10}$ in Single & Double Precision format.

① Convert the no to binary.

$(1259)_{10} = (10011101011)_2$

$(.125)_{10} = (0010)_2$

$(10011101011.0010)_2$

② Normalize the no (Represent in scientific Notation)

Single Precision Formula: $(1.N)\,2^{E-127}$

Double Precision Formula: $(1.N)\,2^{E-1023}$

N – Mantissa
2 – Base
E-127, E-1023 – Exponents
1 – Sign bit

$(100111010011.0010)_2$

Acc to formula

Exponent = $E-127$

But Acc to eq $E=10$

Rule :- Move . to the Right of 1st one.

$$1.00111010011 0010 \times 2^{10}$$

Mantissa

$$E-127 = 10$$

$$E = 10+127$$

$$= (137)_{10}$$

$$= (10001001)_2$$

Single Precision Format :- $(1.N)\,2^{E-127}$

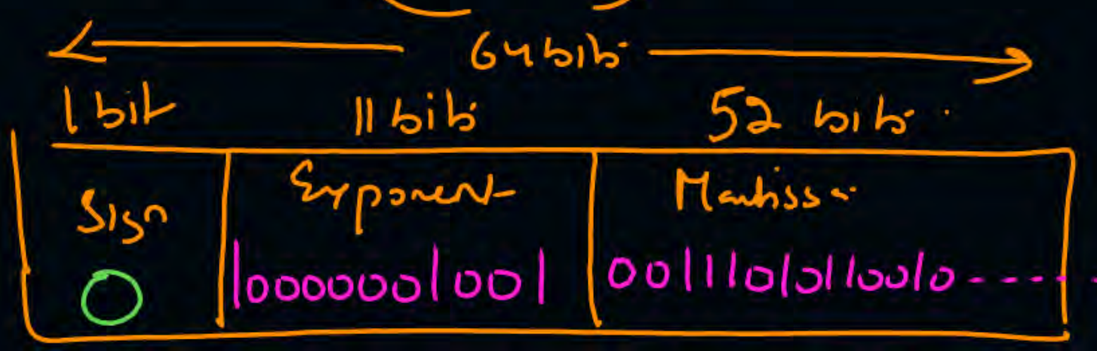
32 bits

Sign 1	Exponent 8	Mantissa 23 bits
0	10001001	00111010011 0010 ----

Attach 0's till 23 bits.

Double Precision

$(1.N)_2$ $E-1023$



$1.\underbrace{00111010110010}_{\text{Mantissa}} \times 2^{(10)}$

Acc to formula $E \Rightarrow E-1023$

Acc to q $E \Rightarrow 10$

$E-1023 = 10$

$E = 10 + 1023$

$= (1033)$

$= (1033)_{10}$

$= (10000001001)_2$

Topic : Floating Point Representation

SINGLE PRECISION	DOUBLE PRECISION
In single precision, 32 bits are used to represent floating-point number.	In double precision, 64 bits are used to represent floating-point number.
This format, also known as FP32, is suitable for calculations that won't be adversely affected by some approximation.	This format, often known as FP64, is suitable to represent values that need a wider range or more exact computations.
It uses 8 bits for exponent.	It uses 11 bits for exponent.
In single precision, 23 bits are used for mantissa.	In double precision, 52 bits are used for mantissa.

Topic : Floating Point Representation

SINGLE PRECISION	DOUBLE PRECISION
Bias number is 127.	Bias number is 1023.
Range of numbers in single precision : $2^{(-126)}$ to $2^{(+127)}$	Range of numbers in double precision : $2^{(-1022)}$ to $2^{(+1023)}$
This is used where precision matters less.	This is used where precision matters more.
It is used for wide representation.	It is used for minimization of approximation.
It is used in simple programs like games.	It is used in complex programs like scientific calculator.

Topic : Floating Point Representation

SINGLE PRECISION	DOUBLE PRECISION
This is called binary 32.	This is called binary 64.
It requires fewer resources as compared to double precision.	It provides more accurate results but at the cost of greater computational power, memory space, and data transfer.
It is less expensive.	The cost incurred using this format does not always justify its use for every computation .

2 Mins Summary

→ Floating pt Rep.

+ Concept

+ Win Q's

THANK YOU

Tuesday
Youtube — 11:00 AM